MTH102

Linear Algebra

2025–2026–I Semester

11 September 2025

Preface

This discusses the content covered in the class, along with some examples, exercises, and problem sets. You are encouraged to go through it. Solving the exercises would greatly benefit in understanding the content of the course. The typos/inaccuracies may be reported to jpsaha@iiserb.ac.in.

Timetable

The classes are scheduled on

- Mondays,
- Wednesdays,
- Fridays

from **11:00am to 11:55am** in L5, LHC. The tutorial is scheduled on

- Thursdays

from **5:30pm to 6:30pm** in L5, LHC.

Office hours

The office hours will be held in **Room no. 108, AB1** during **5:00pm to 6:00pm** on

- Mondays,
- Tuesdays,
- Wednesdays,
- Fridays.

The grading policy is as follows.

Components	Weightage
Quiz	20%
Mid-semester examination	35%
End-semester examination	45%

Calendar

- **Quiz 1** is scheduled on **Friday, 10th October, 2025**, and will be based on the content covered until **Saturday, 4th October, 2025**.
- **Quiz 2** is scheduled on **Friday, 21st November, 2025**, and will be based on the content covered until **Saturday, 15th November, 2025**.
- The academic calendar is available at https://www.iiserb.ac.in/doaa/schedule.

The announcements related to the course will be made on the following Google classroom page.

https://classroom.google.com/u/1/c/ODAwOTg3NDg2Mjg0

The notes can be accessed at the link below (until all the students join the Google classroom).

https://jpsaha.github.io/home/blog/2025/mth102/

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Introduction

Chapter 1. Sets

Definition

A **set** is a well-defined collection of objects. The objects of a set are called its **elements** or **members**.

Examples

- The set of vowels a, e, i, o, u.
- The set consisting of $2, 4, 6, 8, 10$.
- The set consisting of $3, 15, 35, 63, 99$.

Here are further examples.

Examples

- The set of positive integers divisible by 3.
- The set of prime numbers less than 100.
- The set of positive integers which can be expressed as the product of two distinct primes.
- The set of positive integers which can be expressed as the product of two or more distinct primes.
- The set of positive integers which are smaller than 100 and share no common factor with 100.

Remark

Note that the first few sets are described by writing down all its elements, whereas in the latter examples, the sets are described by the rules or properties which determine whether a particular object is an element or not.

Notation

A set is usually denoted by an uppercase letter, for example,

$$A, B, C, X, Y, Z,$$

whereas the lowercase letters are used to denote its elements. For instance, one may use

- a to denote an element of A,
- b to denote an element of B,
- c to denote an element of C,
- x to denote an element of X,
- y to denote an element of Y,
- z to denote an element of Z.

Several special sets are denoted by certain standard symbols. Here are some such examples.

$\mathbb{N}$	the set of positive integers
$\mathbb{Z}$	the set of integers

If x is an element of a set X, then one says that $\boldsymbol{x}$ **belongs to** $\boldsymbol{X}$, and writes

$$x \in X.$$

Note that 2 belongs to $\mathbb{N}$, but $\frac{1}{2}$ does not belong to $\mathbb{N}$. One writes

$$2 \in \mathbb{N}, \frac{1}{2} \notin \mathbb{N}.$$

Notation

There are two ways to write down a set. For example, consider the set of vowels, which is written as

$$V = \{\text{a, e, i, o, u}\}.$$

Note that the elements are seperated by commas and enclosed in braces $\{\}$.

Another way to write down a set is by stating the *rules* or *properties* which determine whether a particular object is an element or not. For example, the set of even positive integers is written as

$$E = \{x \in \mathbb{Z} : x \text{ is even}, x > 0\}.$$

This reads

"E is the set of elements x in $\mathbb{Z}$ such that x is even and $x > 0$".

Example

Note that the set $A = \{3, 15, 35, 63, 99\}$ is equal to

$$\{x \in \mathbb{Z} : x \text{ is equal to } n^2 - 1 \text{ for some } n \in \{2, 4, 6, 8, 10\}\}.$$

Since $1, 2$ are the roots of the polynomial $x^2 - 3x + 2$, it follows that

$$\{x \in \mathbb{N} : x^2 - 3x + 2 = 0\} = \{1, 2\}.$$

Also note that

$$\mathbb{N} = \{1, 2, 3, \ldots\},$$
$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \ldots\}.$$

One uses the symbol $:=$ to indicate that the symbol on the left is being defined by the symbol on the right.

$$\mathbb{Q} := \left\{\frac{m}{n} : m, n \in \mathbb{Z} \text{ and } n \neq 0\right\}.$$

It is called the *set of rational numbers*.

If the number of elements of a set is finite, then it is called a **finite set**. If a set is not finite, it is called an **infinite set**.

Example

The sets

$$\{\text{a, e, i, o, u}\}, \{3, 15, 35, 63, 99\}, \{x \in \mathbb{N} : x^2 - 3x + 2 = 0\}$$

are finite. The sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are infinite.

Exercise 1.1. Determine the elements of the following sets.

- $\{x \in \mathbb{N} : x^2 - 1 = 0\}$.
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}$.

Solution. Note that 1 is the only element of $\mathbb{N}$ satisfying the equation $x^2 - 1 = 0$. This gives

$$\boxed{\{x \in \mathbb{N} : x^2 - 1 = 0\} = \{1\}}.$$

Since $1, -1$ are precisely all the elements of $\mathbb{Z}$ satisfying $x^2 - 1 = 0$, it follows that

$$\boxed{\{x \in \mathbb{Z} : x^2 - 1 = 0\} = \{1, -1\}}.$$

□

§1.1 Basic terminologies

If A, B are sets, and every element of A also belongs to B, then we say that A is a **subset** of B, and write

$$A \subseteq B.$$

If A is not a subset of B, one writes

$$A \nsubseteq B.$$

Example

- Note that every set is a subset of itself.
- Note that $\mathbb{N}$ is a subset of $\mathbb{Z}$, and $\mathbb{Z}$ is a subset of $\mathbb{Q}$. One writes

$$\mathbb{N} \subseteq \mathbb{Z}, \quad \mathbb{Z} \subseteq \mathbb{Q}.$$

- Also note that

$$\mathbb{Z} \nsubseteq \mathbb{N}, \mathbb{Q} \nsubseteq \mathbb{Z}, \mathbb{Q} \nsubseteq \mathbb{N}.$$

If A is a subset of B and A is not equal to B, then A is said to be a *proper subset of* B, and one writes

$$A \subsetneq B.$$

Note that

$$\mathbb{N} \subsetneq \mathbb{Z}, \mathbb{Z} \subsetneq \mathbb{Q}.$$

Exercise 1.2. What does it mean to say that A is not a subset of B?

Solution. Note that A is a subset of B is equivalent to the statement that each element of A lies in B. Hence, A is not a subset of B is equivalent to saying that some element of A does not belong to B. □

Exercise 1.3. Show that if A is a subset of B and B is a subset of C, then A is a subset of C.

Solution. Let x be an element of A. Since A is a subset of B, it follows that x belongs to B. Using that B is a subset of C, we obtain that x belongs to C. Hence, the element x lies in C. This shows that any element x of A belongs to C. This proves that A is a subset of C. □

Exercise 1.4. Show that no element n of $\mathbb{N}$ satisfies $n^4 - 5n^2 + 6 = 0$.

Tip

Try to show that no natural number n satisfies any of the equations

$$n^2 = 2, n^2 = 3.$$

Solution. Note that the polynomial $x^4 - 5x^2 + 6$ factorizes as

$$x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3).$$

Let n be a natural number. If $n = 1$, then $n^2 \neq 2$ and $n^2 \neq 3$ hold. If $n \geq 2$, then $n^2 \geq 4$, and hence, $n^2 \neq 2$ and $n^2 \neq 3$ hold. This shows that no natural number n satisfies

$$(n^2 - 2)(n^2 - 3) = 0,$$

that is, the equation

$$n^4 - 5n^2 + 6 = 0.$$

□

Consider the set

$$A = \{n \in \mathbb{N} : n^4 - 5n^2 + 6 = 0\}.$$

Note that the set A has no elements. Any such set is called the **empty set** or **null set**, and is denoted by the symbol

$$\emptyset.$$

Two sets, A and B, are said to be **equal**[1] if any element of A belongs to B, and any element of B also belongs to A, or equivalently,

$$A \subseteq B \text{ and } B \subseteq A$$

holds. If two sets A, B are not equal, one writes

$$A \neq B.$$

Exercise 1.5. What does it mean to say that two sets A, B are not equal?

Tip

Does Exercise 1.2 help?

Solution. Note that two sets A, B are equal is equivalent to saying that A is a subset of B, and B is a subset of C. Hence, the statement $A \neq B$ is equivalent to saying that A is **not** a subset of B, **or** B is **not** a subset of A, which is equivalent to the statement that some element of A does not belong to B, **or** some element of B does not belong to A. □

§1.2 Operations on sets

Exercise 1.6. Consider the sets

$$\{1, 2, 3, 4\}, \{3, 4, 5, 6\}.$$

Determine the elements common to these sets.

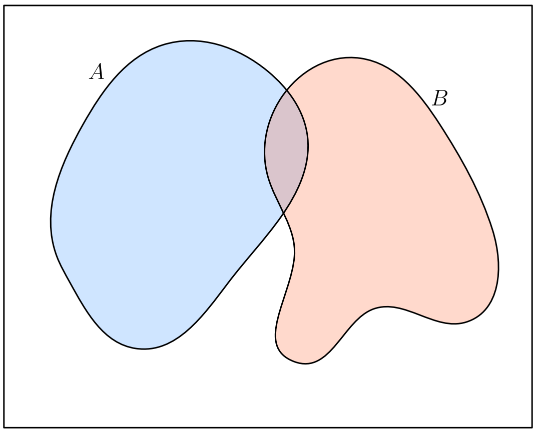

Figure 1: Two sets A and B

[1] *It* may appear obvious! The advantage of putting forth definitions is to set up notations and conventions, so that these are not left to interpretations!

Definition

If A, B are sets, their **union** is denoted by $A \cup B$, which is defined as

$$A \cup B := \{x : x \in A \text{ or } x \in B\}.$$

See Figure 2.

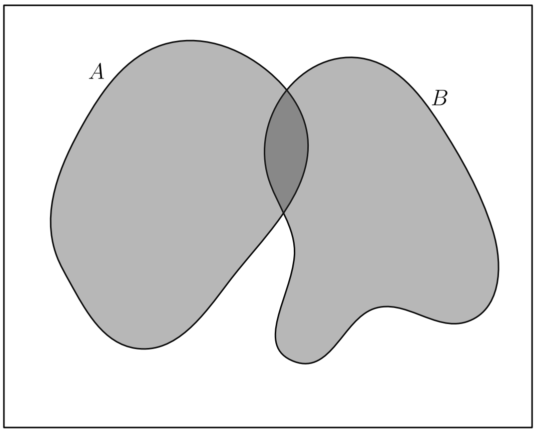

Figure 2: The union of A and B

Exercise 1.7. Determine the union of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Usage of "or" in the inclusive sense

The word "or" is used in the inclusive sense, allowing two or more of the conditions to be satisfied. In common terminology, this inclusive sense is denoted by "and/or".

Exercise 1.8. If A, B are sets, show that

$$A \subseteq A \cup B, B \subseteq A \cup B.$$

Solution. Let x be an element of A. Then note that x lies $A \cup B$. This shows that A is a subset of $A \cup B$. Similarly, it follows that B is a subset of $A \cup B$. □

Exercise 1.9. If A, B are subsets of a set C, then $A \cup B$ is a subset of C.

Solution. Let x be an element of $A \cup B$. It follows that x belongs to A or x belongs to B. If x belongs to A, then using A is a subset of C, we obtain that x lies in C. Further, if x belongs to B, then using that B is a subset of C, we obtain that x lies in C. Combining the above cases, it follows that x lies in C. Consequently, any element of $A \cup B$ is an element of C, or equivalently, $A \cup B$ is a subset of C. □

Exercise 1.10. If A is a set, show that

$$A \cup A = A.$$

Solution. Note that any element x of $A \cup A$ belongs to A or belongs to A, and hence it lies in A. This shows that $A \cup A$ is a subset of A. Further, for any element y of A, it belongs to $A \cup A$. Hence, A is a subset of $A \cup A$. This proves that

$$\boxed{A \cup A = A}.$$

□

Exercise 1.11 (Commutative property). If A, B are sets, show that

$$A \cup B = B \cup A.$$

Solution. Let P, Q be sets, and let x be an element of $P \cup Q$. It follows that x belongs to P or x belongs to Q. This shows that x belongs to Q or x belongs to P. This implies that x is an element of $Q \cup P$. Hence, $P \cup Q$ is a subset of $Q \cup P$ for any two sets P, Q.

Consequently, the set $A \cup B$ is a subset of $B \cup A$, and $B \cup A$ is a subset of $A \cup B$. This shows that

$$\boxed{A \cup B = B \cup A}.$$

□

Definition

If A, B, C are sets, their **union** is denoted by $A \cup B \cup C$, which is defined as

$$A \cup B \cup C := \{x : x \in A \text{ or } x \in B \text{ or } x \in C\}.$$

Exercise 1.12 (Associative property). If A, B, C are sets, show that

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

Use Exercise 1.11 and Exercise 1.12, to show that for any three sets A, B, C, the following sets are equal

$$A \cup B \cup C, A \cup C \cup B, B \cup A \cup C, B \cup C \cup A, C \cup A \cup B, C \cup B \cup A.$$

Definition

If A, B are sets, their **intersection** is denoted by $A \cap B$, which is defined as

$$A \cap B := \{x : x \in A \text{ and } x \in B\}.$$

See Figure 3.

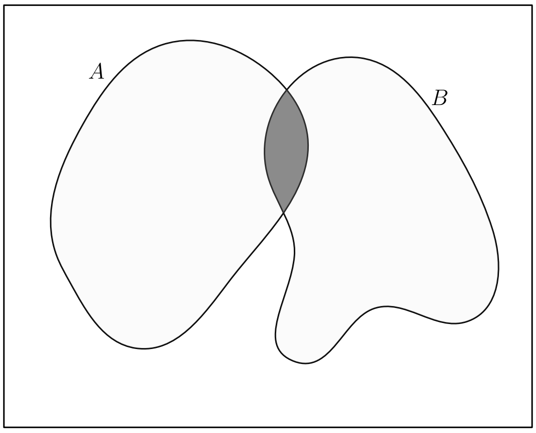

Figure 3: The intersection of A and B

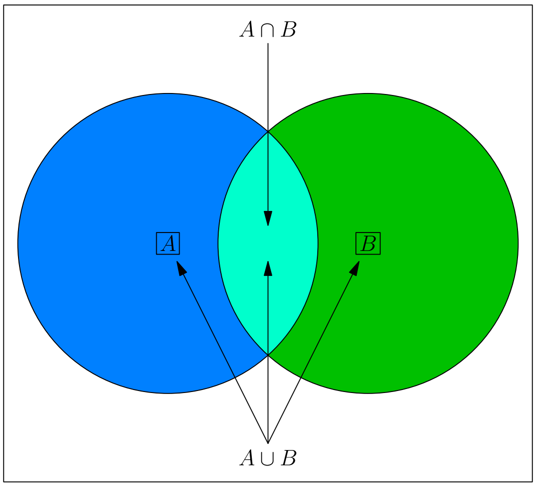

Figure 4: The union and intersection of A and B

Exercise 1.13. Determine the intersection of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.14. If A, B are sets, show that

$$A \cap B \subseteq A, A \cap B \subseteq B.$$

Solution. For any element x of $A \cap B$, it belongs to A and it belongs to B. Hence, $A \cap B$ is a subset of A, and $A \cap B$ is also a subset of B. □

Exercise 1.15. If A, B, C are sets satisfying

$$C \subseteq A, C \subseteq B,$$

then show that

$$C \subseteq A \cap B$$

holds.

Exercise 1.16. Identify the integers among

$$1, 2, 3, 4, \ldots, 20$$

which are divisible by at least one of 2 and 5.

Solution. In the following, the integers among $1, 2, \ldots, 20$ divisible by 2 are marked.

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10,$$

$$11, 12, 13, 14, 15, 16, 17, 18, 19, 20.$$

Note that there 10 integers among $1, 2, \ldots, 20$, which are divisible by 2. The integers among $1, 2, \ldots, 20$ divisible by 5 are marked below.

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10,$$

$$11, 12, 13, 14, 15, 16, 17, 18, 19, 20.$$

Note that there are 4 integers among $1, 2, \ldots, 20$, which are divisible by 5. □

Remark

Note that some integers are counted/marked twice. This is an instance of *double counting*, which refers to the counting of certain element twice. In the above, the integers 10 and 20 are "counted twice". Thus, the number of integers among $1, 2, \ldots, 20$, divisible by 2 or 5, is equal to $10 + 4 - 2 = \boxed{12}$.

Definition

Two sets A, B are said to be **disjoint** if

$$A \cap B = \emptyset.$$

Lemma 1.17 (Inclusion-exclusion principle). *Let X be a set, and A, B be finite subsets of X. Then*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Proof. Note that the set $A \cup B$ is equal to the union of the disjoint sets A and $B \setminus A$. This shows that

$$|A \cup B| = |A| + |B \setminus A|.$$

Also note that the set B is equal to the union of the disjoint subsets $A \cap B$ and $B \setminus A$. This gives

$$|B| = |A \cap B| + |B \setminus A|.$$

It follows that

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}.$$

□

Exercise 1.18 (Inclusion-exclusion principle). Using Lemma 1.17 or otherwise, show that for finite subsets A, B, C of a set X,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Solution. Note that

$$\begin{aligned}
&|A \cup B \cup C| \\
&= |(A \cup B) \cup C| \\
&= |A \cup B| + |C| - |(A \cup B) \cap C| && \text{(by Lemma 1.17)} \\
&= |A \cup B| + |C| - |(A \cap C) \cup (B \cap C)| && \text{(by Exercise 1.23)} \\
&= |A \cup B| + |C| - (|A \cap C| + |B \cap C| - |(A \cap C) \cap (B \cap C)| && \text{(by Lemma 1.17)} \\
&= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |(A \cap C) \cap (B \cap C)| \\
&= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |A \cap C \cap B \cap C| \\
&= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \\
&= |A| + |B| - |A \cap B| + |C| - |A \cap C| - |B \cap C| + |A \cap C \cap B \cap C| && \text{(by Lemma 1.17)} \\
&= |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.
\end{aligned}$$

□

Exercise 1.19. Determine the number of integers among

$$1, 2, 3, 4, \ldots, 100,$$

which are divisible by at least one of $6, 10, 15$.

Solution. Let X denote the set

$$\{1, 2, 3, ..., 100\}.$$

Let A (resp. B, C) denote the set of elements X which are divisible by 6 (resp. 10, 15). Note that

$$|A| = 16, |B| = 10, |C| = 6$$

hold. Observe that $A \cap B$ consists of the elements of X which are divisible by 6 and by 10, that is, divisible by their least common multiple, which is equal to 30. Similarly, $B \cap C$ consists of the elements of X which are divisible by the least common multiple of 10, 15, which is equal to 30. Further, it follows that $C \cap A$ also consists of the multiples of 30 lying between 1 and 100. Moreover, the set $A \cap B \cap C$ consists of the integers between 1 and 100, which are divisible by the least common multiple of $6, 10, 15$, that is, the integer 30. By the inclusion-exclusion principle (Exercise 1.18), we obtain

$$\begin{aligned}
|A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C| \\
&= |A| + |B| + |C| - 3|A \cap B| + |A \cap B \cap C| \\
&= |A| + |B| + |C| - 2|A \cap B| \\
&= 16 + 10 + 6 - 2 \times 3 \\
&= 26.
\end{aligned}$$

Note that the set $A \cup B \cup C$ consists of the integers between 1 and 100 which are divisible by at least one of 6, 10, 15. This shows that there are precisely $\boxed{26}$ such integers. □

Exercise 1.20. If A is a set, show that

$$A \cap A = A.$$

Tip

Does Exercise 1.10 help? If not, what about its solution?

Exercise 1.21 (Commutative property). If A, B are sets, show that

$$A \cap B = B \cap A.$$

Tip

Does Exercise 1.11 help? If not, what about its solution?

Definition

If A, B, C are sets, their **intersection** is denoted by $A \cap B \cap C$, which is defined as

$$A \cap B \cap C := \{x : x \in A \text{ and } x \in B \text{ and } x \in C\}.$$

Exercise 1.22 (Associative property). If A, B, C are sets, show that

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C).$$

Tip

Does Exercise 1.12 help? If not, what about its solution?

Exercise 1.23 (Distributive property). If A, B, C are sets, show that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Tip

Recall when two sets are equal.

Solution. Note that if P is a subset of Q, then we claim that $A \cup P$ is a subset of $A \cup Q$. Indeed, for an element x of $A \cup P$, note that x lies in A, or it lies in P. If x lies in A, then it lies in $A \cup Q$. If x lies in P, it also lies in Q, and hence it lies in $A \cup Q$. This proves the claim. Consequently, $A \cup (B \cap C)$ is a subset of both of the sets

$$A \cup B, A \cup C,$$

and hence, we obtain

$$\boxed{A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)}.$$

Let y be an element of $(A \cup B) \cap (A \cup C)$. Note that y lies in $A \cup B$ and y also lies in $A \cup C$. If y lies in A, then it also lies in $A \cup (B \cap C)$. Thus, it remains to consider the case that y does not lie in A, which we assume from now on. Since y lies in $A \cup B$ and y does not lie in A, it follows that y lies in B. Similarly, it also follows that y lies in C. This shows that y lies in $B \cap C$, and hence it lies in $A \cup (B \cap C)$. This proves that

$$\boxed{(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)}.$$

This establishes that

$$\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}.$$

Let us now establish that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Let x be an element of $A \cap (B \cup C)$. Note that x lies in A and also lies in $B \cup C$. If x lies in B, then x lies in $A \cap B$, and hence x belongs to $(A \cap B) \cup (A \cap C)$. It remains to consider the case that x does not lie in B, which we assume from now on. Since x lies in $B \cup C$ and x does not lie in B, it follows that x lies in C. Using that x lies in A, we obtain that x lies in $A \cap C$, and hence, x lies in $(A \cap B) \cup (A \cap C)$. Combining these cases, we obtain

$$\boxed{A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)}.$$

Note that if P is a subset of Q, then $A \cap P$ is contained in $A \cap Q$. It follows that $A \cap B$ is a subset of $A \cap (B \cup C)$, and $A \cap C$ is also a subset of $A \cap (B \cup C)$. This shows that

$$\boxed{(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)}.$$

This proves that

$$\boxed{A \cap (B \cup C) = (A \cap B) \cup (A \cap C)}.$$

□

Remark

All sets under consideration in a given context, are assumed to be contained in a 'large' fixed set, called the *universal set*. For example, while studying the positive integers (for instance, some sets consisting of the positive integers), the set $\mathbb{N}$ can be taken as the universal set. While studying the integers, the set $\mathbb{Z}$ can be taken as the universal set.

Definition

Let A, B be subsets of a set X. The **complement of A in B** is denoted by $B \setminus A$, and is defined by

$$B \setminus A := \{x \in X : x \in B, x \notin A\}$$

It is also called the **difference of B and A**.

The complement of A in X, is called the *complement of* A, and is denoted by A^c, and is defined to be

$$A^c := X \setminus A.$$

Exercise 1.24. Show that

$$A^c = \{x \in X : x \notin A\},$$

where X denotes the underlying universal set.

Solution. Since $X \setminus A$ is equal to $\{x \in X : x \notin A\}$, it follows that

$$\boxed{A^c = \{x \in X : x \notin A\}}\,.$$

□

Exercise 1.25. Determine the complement of $\{1, 2, 3, 4\}$ in $\{1, 3, 5, 7\}$, and the complement of $\{1, 3, 5, 7\}$ in $\{1, 2, 3, 4\}$.

Exercise 1.26. If A, B are subsets of a set X, show that

$$A^c \cap B = B \setminus A.$$

Solution. Note that

$$\begin{aligned} A^c \cap B &= (X \setminus A) \cap B \\ &= \{x \in X : x \in X \setminus A \text{ and } x \in B\} \\ &= \{x \in X : x \notin A \text{ and } x \in B\} \\ &= \{x \in X : x \in B \text{ and } x \notin A\} \\ &= B \setminus A. \end{aligned}$$

□

Exercise 1.27. If A is a subset of a set X, then show that

$$A \cup A^c = X, A \cap A^c = \emptyset, (A^c)^c = A.$$

Exercise 1.28. If A, B are subsets of a set X, then show that

$$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c.$$

Solution. Note that

$$\begin{aligned}(A \cup B)^c &= X \setminus (A \cup B) \\ &= \{x \in X : x \notin (A \cup B)\} \\ &= \{x \in X : x \notin A \text{ and } x \notin B\} \\ &= \{x \in X : x \in A^c \text{ and } x \in B^c\} \\ &= \{x \in X : x \in A^c \cap B^c\} \\ &= A^c \cap B^c\end{aligned}$$

holds for any subsets A, B of X.

As a consequence, we obtain

$$(A^c \cup B^c)^c = ((A^c)^c \cap (B^c)^c),$$

which yields

$$(A^c \cup B^c)^c = A \cap B.$$

This implies

$$((A^c \cup B^c)^c)^c = (A \cap B)^c,$$

or equivalently,

$$\boxed{(A \cap B)^c = A^c \cup B^c}.$$

□

Exercise 1.29. Show that for any sets A, B, the following are equivalent.

$$\begin{aligned}A &\subseteq B, \\ A \cap B &= A, \\ A \cup B &= B, \\ B^c &\subseteq A^c.\end{aligned}$$

Exercise 1.30. If A, B, C are sets, show that

1. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$,
2. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

Tip

Do the above look familiar in the special case when A is equal to the universal set?

Solution. Let X denote the underlying universal set. Note that

$$\begin{aligned}X \setminus (B \cup C) &= (B \cup C)^c \\ &= B^c \cap C^c \\ &= (X \setminus B) \cap (X \setminus C).\end{aligned}$$

This shows that

$$A \cap (X \setminus (B \cup C)) = A \cap ((X \setminus B) \cap (X \setminus C)),$$

which implies that

$$A \cap (X \setminus (B \cup C)) = (A \cap (X \setminus B)) \cap (A \cap (X \setminus C)).$$

Note that for any subset Y of X, we have

$$\begin{aligned} A \cap (X \setminus Y) &= \{x \in X : x \in A \text{ and } x \in X \setminus Y\} \\ &= \{x \in X : x \in A \text{ and } x \notin Y\} \\ &= A \setminus Y. \end{aligned}$$

Consequently, we obtain

$$\boxed{A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)}\ .$$

Similarly, we obtain

$$\begin{aligned} X \setminus (B \cap C) &= (B \cap C)^c \\ &= B^c \cup C^c \\ &= (X \setminus B) \cup (X \setminus C). \end{aligned}$$

This shows that

$$A \cap (X \setminus (B \cap C)) = A \cap ((X \setminus B) \cup (X \setminus C)).$$

Using the distributive property of intersection over union (see Exercise 1.23), we obtain

$$A \cap (X \setminus (B \cap C)) = (A \cap (X \setminus B)) \cup (A \cap (X \setminus C)).$$

Since for any subset Y of X, the equality

$$A \cap (X \setminus Y) = A \setminus Y$$

holds, we conclude that

$$\boxed{A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)}\ .$$

□

Definition

The **symmetric difference** of two sets A, B, denoted by $A\Delta B$, is defined as

$$A\Delta B := (A \cup B) \setminus (A \cap B).$$

Exercise 1.31. Determine the symmetric difference of $\{1, 2, 3, 4\}$ and $\{1, 3, 5, 7\}$.

Exercise 1.32. If A, B are sets, then show that

$$A\Delta B = B\Delta A,$$
$$A \cup B = (A\Delta B) \cup (A \cap B),$$

and

$$(A\Delta B) \cap (A \cap B) = \emptyset.$$

Solution. Note that

$$\begin{aligned} A\Delta B &= (A \cup B) \setminus (A \cap B) \\ &= (B \cup A) \setminus (B \cap A) \\ &= B\Delta A. \end{aligned}$$

Since $A \cap B$ is a subset of $A \cup B$, we obtain

$$\begin{aligned} A \cup B &= ((A \cup B) \setminus (A \cap B)) \cup (A \cap B) \\ &= (A\Delta B) \cup (A \cap B). \end{aligned}$$

Using $A\Delta B = (A \cup B) \setminus (A \cap B)$, it follows that $A\Delta B$ contains no element of $A \cap B$. This implies that

$$(A\Delta B)\cap(A\cap B)=\emptyset.$$

□

Exercise 1.33. Do there exist three finite sets A, B, C satisfying

$$|A\Delta B|=1, |B\Delta C|=2, |C\Delta A|=4?$$

Solution. Note that for any three sets A, B, C,

$$\begin{aligned} C\setminus A &= (C\setminus A)\cap(B\cup B^c) \\ &= ((C\setminus A)\cap B)\cup((C\setminus A)\cap B^c) \\ &\subseteq (A^c\cap B)\cup(C\cap B^c) \\ &= (B\setminus A)\cup(C\setminus B) \\ &\subseteq (A\Delta B)\cup(B\Delta C) \end{aligned}$$

hold. Hence, for any three sets A, B, C, we also obtain

$$\begin{aligned} A\setminus C &\subseteq (C\Delta B)\cup(B\Delta A) \\ &= (A\Delta B)\cup(B\Delta C). \end{aligned}$$

This shows that for any three sets A, B, C, the union of $A\setminus C, C\setminus A$ is a subset of $(A\Delta B)\cup(B\Delta C)$, that is, $C\Delta A$ is a subset of $(A\Delta B)\cup(B\Delta C)$. If A, B, C are finite sets, it follows that

$$\boxed{|C\Delta A|\leq|(A\Delta B)\cup(B\Delta C)|\leq|A\Delta B|+|B\Delta C|}.$$

This shows that there are no three finite sets A, B, C satisfying the given conditions. □

§1.3 Family of sets

Definition

Let $n\geq 2$ be an integer, and let $A_1, ..., A_n$ be sets. The **union of $A_1, ..., A_n$** is denoted by $A_1\cup\cdots\cup A_n$, and is defined by

$$A_1\cup\cdots\cup A_n := \{x : x \text{ belongs to } A_i \text{ for some } 1\leq i\leq n\}.$$

The **intersection of $A_1, ..., A_n$** is denoted by $A_1\cap\cdots\cap A_n$, and is defined by

$$A_1\cap\cdots\cap A_n := \{x : x \text{ belongs to } A_i \text{ for all } 1\leq i\leq n\}.$$

§1.4 Power set

Example

Note that the subsets of $\{1,2,3\}$ are

$$\begin{gathered} \{1,2,3\}, \\ \{1,2\},\{2,3\},\{3,1\}, \\ \{1\},\{2\},\{3\}, \\ \emptyset. \end{gathered}$$

Definition

Let A be a set. The set of subsets of A is denoted by $\mathcal{P}(A)$. It is called the **power set of A**.

Example

Note that

$$\begin{aligned}
\mathcal{P}(\{1\}) &= \{\emptyset, \{1\}\}, \\
\mathcal{P}(\{x\}) &= \{\emptyset, \{x\}\}, \\
\mathcal{P}(\{x, y\}) &= \{\emptyset, \{x\}, \{y\}, \{x, y\}\}, \\
\mathcal{P}(\{x, y, z\}) &= \{\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{y, z\}, \{z, x\}, \{x, y, z\}\}, \\
\mathcal{P}(\{1, 2, 3\}) &= \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}\}, \\
\mathcal{P}(\{1, 2, 3, 4\}) &= \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}, \\
&\qquad \{4\}, \{1, 4\}, \{2, 4\}, \{3, 4\}, \{1, 2, 4\}, \{2, 3, 4\}, \{1, 3, 4\}, \{1, 2, 3, 4\}\}.
\end{aligned}$$

Remark

Convince yourself that for any integer $n \geq 0$, the power set of a set of size n has size 2^n.

§1.5 Cartesian product of sets

Definition

If A, B are sets, then the **cartesian product** $A \times B$ of A and B is the set of all ordered pairs of the form (a, b) with $a \in A$ and $b \in B$. That is,

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

If $A = \{1, 2\}$ and $B = \{2, 3, 4\}$, then

$$A \times B = \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4)\}.$$

Exercise 1.34. If A, B are finite sets, then show that

$$|A \times B| = |A| \times |B|,$$

where for a set X, its number of elements is denoted by $|X|$.

§1.6 Sets and subsets of real and complex numbers

$$\begin{aligned}
\mathbb{N} &= \text{the set of positive integers} \\
&= \{1, 2, 3, \ldots\}, \\
\mathbb{Z} &= \text{the set of integers} \\
&= \{0, 1, -1, 2, -2, 3, -3, \ldots\}, \\
\mathbb{Q} &= \text{the set of rational numbers} \\
&= \left\{\frac{m}{n} : m \in \mathbb{Z}, n \in \mathbb{N}\right\}, \\
\mathbb{R} &= \text{the set of real numbers}.
\end{aligned}$$

Remark

There are real numbers which are not rational. For instance, the number $\sqrt{2}$ is not rational.

Lemma 1.35. *No rational number x satisfies $x^2 = 2$.*

Proof. On the contrary, let us assume that some rational number x satisfies $x^2 = 2$. Replacing x by $-x$ if necessary, we may and do assume that x is positive. Write $x = \frac{a}{b}$ for some positive integers a, b. Let d denote the greatest common divisor of a, b. Write $a = da_0, b = db_0$ where a_0, b_0 are positive integers. Note that $\left(\frac{a_0}{b_0}\right)^2 = 2$.

Claim

None of a_0, b_0 is divisible by 3.

Proof of the Claim

On the contrary, let us assume that 3 divides at least one of the integers a_0, b_0.

If 3 divides a_0, then 3 divides a_0^2. Using $a_0^2 = 2b_0^2$, it follows that 3 divides b_0. Further, if 3 divides b_0, then 3 divides $2b_0^2$, and hence 3 divides a_0^2. This shows that a_0 is divisible by 3.

In both the cases, it follows that the greatest common divisor of a_0, b_0 is larger than 1. This shows that the greatest common divisor of da_0, db_0 is larger than d. Using $a = da_0, b = db_0$, we obtain that the greatest common divisor of a, b is larger than d, which is impossible. This contradicts the assumption that some rational number x satisfies $x^2 = 2$. This proves the claim.

Claim

If n is an integer not divisible by 3, then n^2 is equal to $3k + 1$ for some integer k, depending on n.

Proof of the Claim

Let q (resp. r) denote the quotient (resp. remainder) obtained upon dividing n by 3. This gives $n = 3q + r$. Note that

$$\begin{aligned} n^2 &= (3q + r)^2 \\ &= 9q^2 + 6qr + r^2 \end{aligned}$$

Since $n = 3q + r$ and 3 does not divide n, it follows that r is equal to 1 or 2. If $r = 1$, then

$$n^2 = 3(3q^2 + 2qr) + 1.$$

If $r = 2$, then

$$n^2 = 3(3q^2 + 2qr) + 4 = 3(3q^2 + 2qr + 1) + 1.$$

This proves the claim.

Using the claims above, it follows that

$$\begin{aligned} a_0^2 &= 3c + 1, \\ b_0^2 &= 3d + 1 \end{aligned}$$

for some integers c, d. This gives

$$3c + 1 = 2(3d + 1).$$

This yields $1 = 3(c - 2d)$, which is impossible.

This completes the proof of the lemma. ☐

Definition

Let a, b be real numbers with $a \leq b$. Then the **closed interval** $[a, b]$ is defined by
$$[a, b] := \{x \in \mathbb{R} : a \leq x \leq b\}.$$
The **open interval** (a, b) is defined by
$$(a, b) := \{x \in \mathbb{R} : a < x < b\}.$$

Exercise 1.36. Determine the cartesian product of $[1, 2]$ and $[3, 4] \cup [5, 6]$.

Note that no real number x satisfies $x^2 = -1$. To "resolve" this, one "introduces"[2] an element i satisfying
$$\boxed{i^2 = -1.}$$
The element i can be multiplied by any real number, and the product can also be added to any real number. This leads to the numbers of the form $a + ib$, where a, b are real numbers. Such numbers are called the **complex numbers**, and the set of such numbers is denoted by $\mathbb{C}$.

Definition

The **set of complex numbers**, is denoted by $\mathbb{C}$, and is defined by
$$\mathbb{C} := \{a + ib : a, b \in \mathbb{R}\}.$$

Exercise 1.37.

Remark

More formally, one defines $\mathbb{C}$, as the following set
$$\{(a, b) : a, b \in \mathbb{R}\},$$
equipped with addition and multiplication defined by
$$(a, b) + (c, d) := (a + c, b + d),$$
$$(a, b) \cdot (c, d) := (ac - bd, ad + bc),$$
for any $(a, b), (c, d)$ with $a, b, c, d \in \mathbb{R}$.

[2]To introduce is to adjoin.

Chapter 2. Induction principles

Example

Show that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

holds for any positive integer n.

Solution

For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Note that $P(1)$ holds. Let k be an element of $\mathbb{N}$, and assume that $P(k)$ holds. Note that

$$\begin{aligned} 1 + 2 + 3 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \quad (\text{using } P(k)) \\ &= (k+1)\left(\frac{k}{2} + 1\right) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n.

Definition

If A is a nonempty subset of $\mathbb{N}$, then an element a_0 of A is called a **least element** of A if $a_0 \leq a$ for all $a \in A$.

Exercise 2.1. If A is a nonempty subset of $\mathbb{N}$, and a_1, a_2 are least elements of A, then show that $a_1 = a_2$.

Solution. Since a_1, a_2 are least elements of A, they belong to A. Using that a_1 is a least element of A, we obtain that $a_1 \leq a_2$. Further, using that a_2 is a least element of A, we obtain that $a_2 \leq a_1$. Combining these two inequalities, it follows that $a_1 = a_2$. □

Remark

By Exercise 2.1, a nonempty subset of $\mathbb{N}$ has at most one least element, to be called **the** least element, if it exists.

Well-ordering principle

Every nonempty subset of $\mathbb{N}$ has a least element.

Principle of mathematical induction

For each $n \in \mathbb{N}$, let $P(n)$ be a statement. Suppose $P(1)$ is true, and that whenever $P(n)$ holds for some $n \geq 1$, the statement $P(n+1)$ holds. Then $P(n)$ is true for all $n \geq 1$.

Principle of strong induction

For each $n \in \mathbb{N}$, let $P(n)$ be a statement. Suppose $P(1)$ is true, and that whenever $P(1), P(2), \ldots, P(n)$ hold for some $n \geq 1$, the statement $P(n+1)$ holds. Then $P(n)$ is true for all $n \geq 1$.

i **Remark**

The above three principles are equivalent.

Exercise 2.2 (Infinitude of primes, by Saidak). Let $a_1 = 2$, and $a_{n+1} = a_n(a_n + 1)$ for any positive integer n. For any $n \in \mathbb{N}$, show that a_n has at least n distinct prime divisors.

Solution. For a positive integer n, let $P(n)$ denote the statement that $a_n \geq 1$ and a_n has at least n distinct prime divisors.

Since $a_1 = 2$, it follows that $P(1)$ is true.

Let n be a positive integer such that $P(n)$ holds. This gives that $a_n \geq 1$, and hence $a_n + 1 \geq 2$. Also note that the integers $a_n, a_n + 1$ have no common prime divisor, since any of their common divisors divides their difference, which is equal to 1. Hence, no prime divisor of a_n is a prime divisor of $a_n + 1$. By the induction hypothesis, a_n has at least n distinct prime divisors. Since $a_n + 1 \geq 2$, it admits at least one prime divisor. It follows that the product $a_n(a_n + 1)$ has at least $n + 1$ distinct prime divisors. Also note that $a_n(a_n + 1) \geq 2$ holds. This shows that $P(n + 1)$ holds.

By induction, the statement $P(n)$ holds for all $n \in \mathbb{N}$. In particular, a_n has at least n distinct prime divisors. □

Exercise 2.3. Show that there are infinitely many prime numbers of the form $4m + 3$, where m is a nonnegative integer.

Solution. Le $a_1, a_2, \ldots$ be a sequence of positive integers defined by

$$a_1 := 7,$$
$$a_{n+1} := a_n(4a_n + 3)$$

for all $n \in \mathbb{N}$. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that a_n is a positive integer, 3 does not divide a_n, and a_n has at least n distinct prime divisors of the form $4m + 3$.

Since $a_1 = 7$, it follows that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Since a_k is a positive integer, it follows that $4a_k + 3$ is an odd positive integer ≥ 3. Hence it has[3] a prime divisor of the form $4m + 3$. Since 3 does not divide a_k, it follows[4] that 3 does not divide $4a_k + 3$. Also note that any common factor of the integers a_k and $4a_k + 3$ divides

$$(4a_k + 3) - 4a_k,$$

[3] How does it follow? Prove it by induction that for any positive integer n, the product of any integers of the form $4m + 1$ is also of the form $4m + 1$.

[4] Why?

which is equal to 3. Since 3 does not divide a_k, we obtain that the integers a_k and $4a_k + 3$ have no common prime divisor. Hence, any prime divisor of $4a_k + 3$ of the form $4m + 3$ is different from the prime divisors of a_k. This shows that a_{k+1} is a positive integer, 3 does not divide a_{k+1}, and a_{k+1} has at least $k + 1$ distinct prime divisors of the form $4m + 3$. By induction, it follows that for any $n \in \mathbb{N}$, the integer a_n has n distinct prime divisors of the form $4m + 3$.

This shows that there are infinitely many prime numbers of the form $4m + 3$. □

Exercise 2.4. Show that

$$2^{2^n} - 1$$

has at least n distinct prime divisors for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that $2^{2^n} - 1$ has at least n distinct prime divisors.

Since $2^{2^1} - 1 = 3$, it follows that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$2^{2^{k+1}} - 1 = \left(2^{2^k} - 1\right)\left(2^{2^k} + 1\right).$$

Also note that the integers $2^{2^k} - 1$ and $2^{2^k} + 1$ have no common prime divisors, since these are odd integers, and any of their common divisors divides their difference, which is equal to 2. By the induction hypothesis, the integer $2^{2^k} - 1$ has at least k distinct prime divisors. Since $2^{2^k} + 1 \geq 5$, it admits at least one prime divisor. It follows that the product $\left(2^{2^k} - 1\right)\left(2^{2^k} + 1\right)$ has at least $k + 1$ distinct prime divisors. This shows that $P(k + 1)$ holds.

By induction, the statement $P(n)$ holds for all $n \in \mathbb{N}$. □

Remark

Note that it is false that

$$2^{2^n} + 1$$

has at least n distinct prime divisors for all $n \in \mathbb{N}$. Indeed, the integer $2^{2^2} + 1 = 17$ is a prime.

Exercise 2.5. Show that every positive integer greater than 1 is a prime or is a product of prime numbers.

Exercise 2.6. Let S be a finite set of size n. Show that the number of subsets of S is 2^n.

Solution. For a nonnegative integer n, let $P(n)$ denote the statement that if A is a set with n elements, then the number of subsets of A is 2^n.

Since the empty set has exactly one subset, namely itself, it follows that $P(0)$ holds.

Let k be a nonnegative integer such that $P(k)$ holds, that is, for **any set** with k elements, the number of its subsets is 2^k. We show that $P(k + 1)$ holds.

Let A be a set with $k + 1$ elements. Choose an element $a \in S$, and let $B = A \setminus \{a\}$. Note that B has exactly k elements. By the induction hypothesis, the number of subsets of B is 2^k. Any subset of A either contains a or does not contain a. The number of subsets of A that do not contain a is equal to the number of subsets of B, which is 2^k.

Note that any subset of A that contains a is of the form $C \cup \{a\}$, where C is a subset of B. Further, different subsets of B give rise to different subsets of A that contain a. That is, if C_1, C_2 are different subsets of B, then $C_1 \cup \{a\} \neq C_2 \cup \{a\}$. This shows that the number of subsets of A that contain a is also equal to the number of subsets of B, which is again 2^k. It follows that the total number of subsets of A is equal to

$$2^k + 2^k = 2^{k+1}.$$

This shows that $P(k+1)$ holds.

By induction, it follows that for any nonnegative integer n, if A is a set of size n, then the number of subsets of A is 2^n. $\square$

Exercise 2.7. Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.8. Show that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.9. Show that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
&\frac{1}{1\cdot 2\cdot 3}+\frac{1}{2\cdot 3\cdot 4}+\cdots+\frac{1}{k(k+1)(k+2)}+\frac{1}{(k+1)(k+2)(k+3)}\\
&=\frac{k(k+3)}{4(k+1)(k+2)}+\frac{1}{(k+1)(k+2)(k+3)} \qquad \text{(using } P(k))\\
&=\frac{k(k+3)^2+4}{4(k+1)(k+2)(k+3)}\\
&=\frac{(k+1)(k+3)^2-((k+3)^2-4)}{4(k+1)(k+2)(k+3)}\\
&=\frac{(k+1)(k+3)^2-((k+3)^2-2^2)}{4(k+1)(k+2)(k+3)}\\
&=\frac{(k+1)(k+3)^2-(k+1)(k+5)}{4(k+1)(k+2)(k+3)}\\
&=\frac{(k+1)((k+3)^2-(k+5))}{4(k+1)(k+2)(k+3)}\\
&=\frac{k^2+6k+9-(k+5)}{4(k+2)(k+3)}\\
&=\frac{k^2+5k+4}{4(k+2)(k+3)}\\
&=\frac{(k+1)((k+1)+3)}{4((k+1)+1)((k+1)+2)}.
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.10. Show that

$$3+11+\cdots+(8n-5)=4n^2-n$$

for all $n\in\mathbb{N}$.

Exercise 2.11. Show that

$$1^2+3^2+\cdots+(2n-1)^2=\frac{4n^3-n}{3}$$

for all $n\in\mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$1^2+3^2+\cdots+(2n-1)^2=\frac{4n^3-n}{3}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$
\begin{aligned}
&1^2 + 3^2 + \cdots + (2k-1)^2 + (2(k+1)-1)^2 \\
&= \frac{4k^3 - k}{3} + (2k+1)^2 && \text{(using } P(k)\text{)} \\
&= \frac{4k^3 - k + 3(2k+1)^2}{3} \\
&= \frac{4k^3 - k + 12k^2 + 12k + 3}{3} \\
&= \frac{4k^3 + 12k^2 + 11k + 3}{3} \\
&= \frac{4(k+1)^3 - (k+1)}{3}.
\end{aligned}
$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.12. Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1} n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1} n^2 = (-1)^{n+1} \frac{n(n+1)}{2}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$
\begin{aligned}
&1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{k+1} k^2 + (-1)^{(k+1)+1} (k+1)^2 \\
&= (-1)^{k+1} \frac{k(k+1)}{2} + (-1)^{k+2} (k+1)^2 && \text{(using } P(k)\text{)} \\
&= (-1)^{k+1} \left(\frac{k(k+1)}{2} - (k+1)^2 \right) \\
&= (-1)^{k+1} \left(\frac{k(k+1) - 2(k+1)^2}{2} \right) \\
&= (-1)^{k+1} \left(\frac{-(k+1)(k+2)}{2} \right) \\
&= (-1)^{(k+1)+1} \frac{(k+1)((k+1)+1)}{2}.
\end{aligned}
$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.13. Show that

$$
\begin{aligned}
1^2 + 2^2 + 3^2 + \cdots + n^2 &= \frac{n(n+1)(2n+1)}{6}, \\
1^3 + 2^3 + 3^3 + \cdots + n^3 &= \left(\frac{n(n+1)}{2} \right)^2
\end{aligned}
$$

hold for any positive integer n.

Exercise 2.14. Show that
$$\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{2n-1}-\frac{1}{2n}$$
for all $n \in \mathbb{N}$.

Exercise 2.15. Show that n^3+5n is divisible by 6 for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that n^3+5n is divisible by 6.

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that
$$\begin{aligned}(k+1)^3+5(k+1) &= (k^3+3k^2+3k+1)+(5k+5)\\ &= (k^3+5k)+(3k^2+3k+6).\end{aligned}$$
By the induction hypothesis, the integer k^3+5k is divisible by 6. Also note that $3k^2+3k+6=3(k^2+k+2)$ is divisible by 3. Further, one of the integers $k, k+1$ is even, and hence k^2+k is even. This gives that k^2+k+2 is even, and hence $3(k^2+k+2)$ is divisible by 2. It follows that $3(k^2+k+2)$ is divisible by 6. This shows that $(k+1)^3+5(k+1)$ is divisible by 6. This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.16. Show that $5^{2n}-1$ is divisible by 8 for all $n \in \mathbb{N}$.

Solution. For a positive integer n, let $P(n)$ denote the statement that $5^{2n}-1$ is divisible by 8.

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that
$$\begin{aligned}5^{2(k+1)}-1 &= 5^{2k+2}-1\\ &= 5^2\cdot 5^{2k}-1\\ &= 25\cdot 5^{2k}-1\\ &= (24+1)\cdot 5^{2k}-1\\ &= 24\cdot 5^{2k}+(5^{2k}-1).\end{aligned}$$
By the induction hypothesis, the integer $5^{2k}-1$ is divisible by 8. Also note that $24\cdot 5^{2k}$ is divisible by 8. This shows that $5^{2(k+1)}-1$ is divisible by 8. This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.17. Show that 5^n-4n-1 is divisible by 16 for all $n \in \mathbb{N}$.

Exercise 2.18. Show that 6^n-5n-1 is divisible by 25 for all $n \in \mathbb{N}$.

Exercise 2.19. Show that $n^3+(n+1)^3+(n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.

Exercise 2.20. Show that
$$\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\cdots+\frac{1}{\sqrt{n}}>\sqrt{n}$$
for all $n \in \mathbb{N}$ satisfying $n>1$.

Solution. For a positive integer n, let $P(n)$ denote the statement that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

holds if $n > 1$.

Note that $P(2)$ holds, since

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} = 1 + \frac{\sqrt{2}}{2} > \sqrt{2}.$$

Let k be a positive integer such that $k \geq 2$ and $P(k)$ holds. Note that

$$\begin{aligned} &\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \\ &> \sqrt{k} + \frac{1}{\sqrt{k+1}} && \text{(using } P(k)) \\ &\geq \sqrt{k+1}, \end{aligned}$$

where the last inequality follows since

$$\begin{aligned} k - \left(\sqrt{k+1} - \frac{1}{\sqrt{k+1}}\right)^2 &= k - (k+1) + 2 - \frac{1}{k+1} \\ &= 1 - \frac{1}{k+1} \\ &> 0. \end{aligned}$$

This shows that if k is a positive integer satisfying $k \geq 2$ and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n satisfying $n \geq 2$.

□

Exercise 2.21. Show that $3^n \geq n^2$ for all $n \in \mathbb{N}$.

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that $3^n \geq n^2$. Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned} (k+1)^2 &= k^2 + 2k + 1 \\ &\leq 3^k + 2k + 1 && \text{(using } P(k)) \\ &\leq 3^k + 3^k + 3^k && \text{(using } k \geq 1) \\ &= 3 \cdot 3^k \\ &= 3^{k+1}. \end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. □

Exercise 2.22. Show that $n! > 2^n$ for all $n \in \mathbb{N}$ satisfying $n \geq 4$.

Solution. For a positive integer n, let $P(n)$ denote the statement that $n! > 2^n$.

Note that $P(4)$ holds, since $4! = 24 > 16 = 2^4$.

Let k be a positive integer such that $k \geq 4$ and $P(k)$ holds. Note that

$$\begin{aligned} (k+1)! &= (k+1)k! \\ &\geq (k+1) \cdot 2^k && \text{(using } P(k)) \\ &> 2 \cdot 2^k && \text{(using } k \geq 4) \\ &= 2^{k+1}. \end{aligned}$$

This shows that if k is a positive integer satisfying $k \geq 4$ and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n satisfying $n \geq 4$.

□

Exercise 2.23. Show that

$$\left(1+\frac{1}{1^3}\right)\left(1+\frac{1}{2^3}\right)\left(1+\frac{1}{3^3}\right)\cdots\left(1+\frac{1}{n^3}\right) < 3$$

holds for all $n \in \mathbb{N}$.

Remark

Does taking $P(n)$ to be the statement that

$$\left(1+\frac{1}{1^3}\right)\left(1+\frac{1}{2^3}\right)\left(1+\frac{1}{3^3}\right)\cdots\left(1+\frac{1}{n^3}\right) < 3$$

help?

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$\left(1+\frac{1}{1^3}\right)\left(1+\frac{1}{2^3}\right)\left(1+\frac{1}{3^3}\right)\cdots\left(1+\frac{1}{n^3}\right) \leq 3 - \frac{1}{n}.$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
&\left(1+\frac{1}{1^3}\right)\left(1+\frac{1}{2^3}\right)\left(1+\frac{1}{3^3}\right)\cdots\left(1+\frac{1}{k^3}\right)\left(1+\frac{1}{(k+1)^3}\right)\\
&\leq \left(3-\frac{1}{k}\right)\left(1+\frac{1}{(k+1)^3}\right)\\
&= 3 - \frac{1}{k} + \frac{3}{(k+1)^3} - \frac{1}{k(k+1)^3}\\
&= 3 - \frac{1}{k+1} + \frac{1}{k+1} - \frac{1}{k} + \frac{3}{(k+1)^3} - \frac{1}{k(k+1)^3}\\
&= 3 - \frac{1}{k+1} - \frac{1}{k(k+1)} + \frac{3}{(k+1)^3} - \frac{1}{k(k+1)^3}\\
&= 3 - \frac{1}{k+1} - \frac{(k+1)^2 - 3k + 1}{k(k+1)^3}\\
&= 3 - \frac{1}{k+1} - \frac{k^2 - k + 2}{k(k+1)^3}\\
&= 3 - \frac{1}{k+1} - \frac{k(k-1) + 2}{k(k+1)^3}\\
&< 3 - \frac{1}{k+1}
\end{aligned}$$

holds. This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. This implies the given inequality for all $n \in \mathbb{N}$.

□

Exercise 2.24. Show that

$$1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{n^3} < \frac{3}{2}$$

for all $n \in \mathbb{N}$.

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{n^3} \leq \frac{3}{2} - \frac{1}{2n^2}.$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
& 1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{k^3} + \frac{1}{(k+1)^3} \\
& \leq \left(\frac{3}{2} - \frac{1}{2k^2}\right) + \frac{1}{(k+1)^3} && \text{(using } P(k)\text{)} \\
& = \frac{3}{2} - \frac{1}{2k^2} + \frac{1}{(k+1)^3} \\
& = \frac{3}{2} - \frac{1}{2(k+1)^2} + \frac{1}{2(k+1)^2} - \frac{1}{2k^2} + \frac{1}{(k+1)^3} \\
& = \frac{3}{2} - \frac{1}{2(k+1)^2} - \frac{(k+1)^3 - k^2(k+1) - 2k^2}{2k^2(k+1)^3} \\
& = \frac{3}{2} - \frac{1}{2(k+1)^2} - \frac{k^3 + 3k^2 + 3k + 1 - k^3 - k^2 - 2k^2}{2k^2(k+1)^3} \\
& = \frac{3}{2} - \frac{1}{2(k+1)^2} - \frac{3k+1}{2k^2(k+1)^3} \\
& < \frac{3}{2} - \frac{1}{2(k+1)^2}.
\end{aligned}$$

This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. This implies the given inequality for all $n \in \mathbb{N}$.

$\square$

Exercise 2.25. Show that

$$\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{2^2}\right)\left(1 + \frac{1}{2^3}\right)\cdots\left(1 + \frac{1}{2^n}\right) < \frac{5}{2}$$

holds for all $n \in \mathbb{N}$.

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{2^2}\right)\left(1 + \frac{1}{2^3}\right)\cdots\left(1 + \frac{1}{2^n}\right) \leq \frac{5}{2}\left(1 - \frac{2}{2^{n+1} + 1}\right).$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
&\left(1+\frac{1}{2}\right)\left(1+\frac{1}{2^2}\right)\left(1+\frac{1}{2^3}\right)\cdots\left(1+\frac{1}{2^k}\right)\left(1+\frac{1}{2^{k+1}}\right)\\
&\leq \frac{5}{2}\left(1-\frac{2}{2^{k+1}+1}\right)\left(1+\frac{1}{2^{k+1}}\right)\\
&\leq \frac{5}{2}\times\frac{2^{k+1}-1}{2^{k+1}}\\
&\leq \frac{5}{2}\left(1-\frac{1}{2^{k+1}}\right)\\
&\leq \frac{5}{2}\left(1-\frac{1}{2^{k+2}+1}\right).
\end{aligned}$$

This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n. This implies the given inequality for all $n \in \mathbb{N}$. $\square$

Warning

It is important to verify the base case, that is, verifying that $P(1)$ holds, even if it is trivial. For example, if we define $P(n)$ to be the statement that

$$n = n + 1,$$

then $P(n)$ implies $P(n+1)$ for any $n \in \mathbb{N}$. However, $P(1)$ does not hold. Another example is to take $Q(n)$ to be the statement that $2n+1$ is even, then $Q(n)$ implies $Q(n+1)$ for any $n \in \mathbb{N}$. However, $Q(1)$ does not hold.

Here are a few exercises where strong induction is useful.

Exercise 2.26. Let the integers $x_1, x_2, \ldots$ be defined by

$$\begin{aligned}
x_1 &:= 1,\\
x_2 &:= 2,\\
x_{n+2} &:= \frac{1}{2}(x_n + x_{n+1})
\end{aligned}$$

for all $n \in \mathbb{N}$. Show that $1 \leq x_n \leq 2$ for all $n \in \mathbb{N}$.

Tip

For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that $1 \leq x_m \leq 2$ for all $m \in \mathbb{N}$ satisfying $m \leq n$.

Exercise 2.27. The Fibonacci sequence $F_0, F_1, F_2, \ldots$ is defined by

$$\begin{aligned}
F_0 &:= 0,\\
F_1 &:= 1,\\
F_2 &:= 1,\\
F_{n+2} &:= F_n + F_{n+1}
\end{aligned}$$

for all $n \in \mathbb{N}$. Show that

$$F_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

for all $n \in \mathbb{N} \cup \{0\}$.

Exercise 2.28. Show that

$$F_0^2 + F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}$$

for all $n \in \mathbb{N} \cup \{0\}$.